

An Abstract, Guarded Categorical Model of Gradual Typing

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The work of Giovannini et al. [?] constructs a denotational semantics for a simple gradually-typed lambda calculus using guarded type theory. Many of the aspects of the model construction are in fact independent of the details of the particular language being modeled, e.g., the specific types and effects in the language. In this work, we seek to abstract over these details and present an abstract categorical framework for constructing models of gradual typing in guarded type theory.

With this abstract framework in hand, we then define concrete semantics for gradually-typed languages with simple effects.

0.1 Why an Abstract Model?

The model developed by Giovannini et al. suffers from the limitation that it is a single model for a quite bare-bones gradually-typed language. Real-world languages can include effects, polymorphism, dependent types, etc. The goals of an abstract model are twofold:

1. To promote a better understanding of the essence of any guarded relational semantics of gradual typing independent of the particular language features being modeled.
2. To allow for developers of languages to establish graduality for their language by instantiating our abstract framework.

0.2 Challenges

Prior work by New and Licata [?] defines a categorical model for gradual typing that serves as the basis for denotational semantics using *classical* domain theory. Their model is based on *double categories* [?]. Unfortunately, as Giovannini et al. point out, adapting this model to the guarded setting poses significant challenges, and much of their work consists of defining a version of the model that works in guarded type theory.

Inspired by the solution of Giovannini et al., we refine the notion of abstract categorical model of gradual typing into two novel classes of model suitable for the guarded setting. We call these *extensional* and *intensional* models.

0.3 Outline

This work is organized as follows.

- We begin in Section 1 with an overview of the existing abstract double-categorical semantics for gradual typing by New and Licata.
- We review the difficulties in adapting the New-Licata denotational model to the guarded setting.
- We motivate the refinement of the notion of abstract model into extensional and intensional models and present their definitions in full.
- We discuss the construction of an extensional model of gradual typing from a suitable “base” double category of effectful functions and relations.

- With the constructions in hand, we then instantiate the abstract framework to define concrete models of gradual typing capable of modeling languages with simple effects.

0.4 Ambient Guarded Metalanguage

Because guarded type theory plays a central role in the construction of the concrete model in prior work, it is natural to expect that our abstract framework should to some degree incorporate aspects of guarded type theory. To that end, we demand the following of the categories involved in the description of our abstract model:

1. There is an applicative functor $\triangleright : \mathcal{C} \rightarrow \mathcal{C}$.
2. There is a morphism $\text{next} : X \rightarrow \triangleright X$ for any object X .
3. For any morphism f of the form $\triangleright X \rightarrow X$, there is a unique morphism $\text{fix}f : 1 \rightarrow X$ such that $\text{fix}f = f \circ \text{next} \circ \text{fix}f$.

1 Double Categorical Semantics of Graduality

We begin with a review of abstract categorical models for gradual typing described in prior work. New and Licata [?] modeled the graduality and type casts for call-by-name gradual typing using *double categories*, which are defined to be categories internal to the category of categories. That is, a double category $\mathcal{C}$ consists of a category $\mathcal{C}_f$ of “objects and function morphisms” and a category $\mathcal{C}_{sq}$ of “relation morphisms and squares” with functors (reflexive relation) $r : \mathcal{C}_f \rightarrow \mathcal{C}_{sq}$ and (source and target) $s, t : \mathcal{C}_{sq} \rightarrow \mathcal{C}_f$ satisfying $sr = tr = \text{id}$ as well as a composition operation $c : \mathcal{C}_{sq} \times_{s,t} \mathcal{C}_{sq} \rightarrow \mathcal{C}_{sq}$ respecting source and target. This models an abstract notion of functions and relations. For notation, we write function morphisms as $f : A \rightarrow B$ and relation morphisms as $c : A \multimap B$ where $c \in \mathcal{C}_{sq}$ and $s(c) = A$ and $t(c) = B$. Finally, a morphism α from c to d with $s(\alpha) = f$ and $t(\alpha) = g$ is visualized as

$$\begin{array}{ccc} A & \multimap & B \\ f \downarrow & & \downarrow g \\ A' & \multimap & B' \end{array}$$

Such a morphism is thought of as an abstraction of the notion of relatedness of functions: functions take related inputs to related outputs. The composition operations and functoriality give us a notion of composition of relations as well as functions and vertical and horizontal composition of squares. In this work we will be chiefly interested in *locally thin* double categories, that is, double categories where there is at most one square for any f, c, g, d . In this case we use the notation $f \sqsubseteq_d^c g$ to mean that a square like the above exists.

New, Licata and Ahmed [?] extended the axiomatic syntax to call-by-push-value but did not analyze the structure categorically. We fill in this missing analysis now, extending the double categorical semantics given in [?] from cartesian closed categories to CBPV models.

We begin by reviewing the definition of a CBPV model:

Definition 1.1. A CBPV model $\mathcal{M}$ consists of:

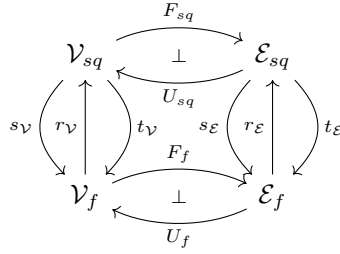
- A cartesian-closed category $\mathcal{V}$ of values and pure morphisms.
- A category $\mathcal{E}$ of computations and linear morphisms.
- Functors $F : \mathcal{V} \rightarrow \mathcal{E}$ and $U : \mathcal{E} \rightarrow \mathcal{V}$ with $F \dashv U$.
- An action of $\mathcal{V}$ on $\mathcal{E}$, i.e., a functor $\rightarrow : \mathcal{V}^{op} \times \mathcal{E} \rightarrow \mathcal{E}$ satisfying the usual coherence laws, and such that $U(A \rightarrow B)$ is isomorphic to $A \Rightarrow UB$.

A *strict homomorphism of CBPV models* from $(\mathcal{V}, \mathcal{E}, \dots)$ to $(\mathcal{V}', \mathcal{E}', \dots)$ consists of functors $G_v : \mathcal{V} \rightarrow \mathcal{V}'$ and $G_e : \mathcal{E} \rightarrow \mathcal{E}'$ that strictly preserve all CBPV constructions; see the appendix for a more detailed definition. We call this a *strict morphism* in contrast to a *lax* morphism, which only preserves CBPV constructions up to transformation.

A model of the congruence rules of the CBPV system can be given by a locally thin “double CBPV model”, which we define below:

Definition 1.2. A *double CBPV model* is a category internal to the category of CBPV models and *strict* homomorphisms of CBPV models¹.

Part of the data of a double CBPV model can be visualized as follows:



The diagram above makes it clear that there are two useful ways of viewing a double CBPV model $\mathcal{M}$. First, we can view $\mathcal{M}$ as pair of CBPV models, one forming the objects and vertical morphisms (the bottom of the diagram) and the other forming the horizontal morphisms and the squares (the top of the diagram). Alternatively, we can view $\mathcal{M}$ as a pair of double categories, one for the values (the left of the diagram), and one for the computations (the right of the diagram). For the latter viewpoint, we will often write $\mathcal{C}$ to stand for an unspecified such double category associated with a double CBPV model.

Given a double CBPV model, we can interpret the syntax of gradual typing as follows. Type precision $A \sqsubseteq A'$ is interpreted as a relation morphism $c_A : A \multimap A'$ in $\mathcal{V}_{sq}$, and term precision $\Gamma \sqsubseteq \Gamma' \vdash M \sqsubseteq M' : A \sqsubseteq A'$ is interpreted as a square $M \sqsubseteq_{U F c_A}^{\text{cf}} M'$. The fact that t, r and the composition are all given by strict CBPV homomorphisms says that all the type constructors lift to precision (monotonicity of type constructors) as well as all term constructors (congruence). Further, r and composition being strict homomorphisms implies that all type constructors strictly preserve the identity relation (identity extension) and composition.

Next, to model type casts, the New-Licata model further requires that every value relation $c : A \multimap A'$ is *left representable* by a function $u_c : A \rightarrow A'$ and every computation relation $d : B \multimap B'$ is *right representable* by a function $d_c : B' \rightarrow B$. In a locally thin double category, left and right representability are defined as follows:

Definition 1.3. A relation $R : A \multimap B$ is *left representable* by $f : A \rightarrow B$ if $f \sqsubseteq_{r(B)}^R \text{id}$ and $\text{id} \sqsubseteq_R^{r(A)} f$.

Dually, $R : A \multimap B$ is *right representable* by $g : B \rightarrow A$ if $\text{id} \sqsubseteq_{r(A)}^R g$ and $g \sqsubseteq_R^{r(B)} \text{id}$.

These squares are sufficient to model the UpL/UpR/DnL/DnR rules for casts. Additionally, since representable morphisms compose, the compositionality of casts comes for free. However, the retraction property must be added as an additional axiom to the model. To model the error being the least element, the New-Licata model adds the requirement that $\text{id} \circ ! \sqsubseteq_{r(U_B)}^{r(A)} f$ holds for all $f : \mathcal{V}(A, B)$. Finally, the dynamic type can be modeled as an arbitrary value type D with arbitrary relations $\text{Nat} \multimap D$ and $D \multimap D \multimap D$ and $D \times D \multimap D$ (or whatever basic type cases are required).

¹it may be possible to also define this as a notion of CBPV model internal to some structured 2-category of categories, but we are not aware of any such definition of an internal CBPV model

2 Adapting the Notion of Model to Guarded Type Theory

2.1 Challenges in Adapting the New-Licata Model

While the original double categorical semantics introduced in Section 1 can be satisfied with classical domain theoretic models as demonstrated by New and Licata [?], Giovannini et al. [?] show that the New-Licata model cannot be directly adapted to work in the guarded setting.

Key to understanding the issue is the fact that the equation for the dynamic type involves a later, which means that when inspecting an element of the dynamic type which is a function, an *observable step* is required in the semantics. This means that downcasts from the dynamic type may take an observable step, and thus *any* cast may take steps as it may involve a downcast from the dynamic type functorially.

The issue in adapting the New-Licata model to the guarded setting is related to the representability condition on the relations. At a high level, the problem arises as a result of a tension between *transitive reasoning* and *step-insensitivity*. Transitive reasoning is desirable because it allows us to derive the cast rules needed for graduality from the simpler, compositional versions of the rules. This is crucial because the original rules quantify over an additional, arbitrary type precision derivation and are therefore not syntax-independent. On the other hand, the graduality property, as axiomatized by the representability of type precision relations, is fundamentally a step-insensitive property. The casts, which may involve observable steps, must be related to the identity function, which does not step. The relation modeling term precision must be able to account for this mismatch.

However, Giovannini et al. show that transitivity and unrestricted step-insensitivity (plus a third minor condition) are incompatible as properties of a relation in guarded type theory (see their Theorem 4.1 for a precise statement). The solution adopted by Giovannini et al. is to work with a *lock-step error ordering* that is transitive, and then restore a small amount of step-insensitivity where needed through explicit synchronizing terms they call *perturbations*. The perturbations mimic the stepping behavior of casts, but are otherwise equivalent to the identity, a property they formalize via a notion of *weak bisimilarity*. The authors relax the definition of representability to allow for casts to be lock-step related to perturbations rather than only to the identity function. Finally, to define the denotation of term precision, the authors take the closure of the lock-step ordering under weak bisimilarity on both sides:

$$M \tilde{\sqsubseteq} N := \exists f, g. M \approx f \sqsubseteq g \approx N,$$

where $\approx$ is weak bisimilarity and $\sqsubseteq$ is the lock-step ordering. The idea is that two terms are related if they can be synchronized in some number of steps to two terms that are *lock-step* related. By separating term precision into a combination of weak bisimilarity and lock-step reasoning, and using syntactic perturbations as a means to explicitly synchronize terms as necessary, the authors achieve compositional reasoning while still accommodating the steps taken by casts.

2.2 Intensional Models of Gradual Typing

An intensional model of gradual typing is an abstraction of working in a setting where the ordering on terms is transitive but requires terms to be in lock-step. To aid in understanding the definition of an intensional model, we break the definition into five steps, with each one building on the previous. The last step will be the definition we have in mind when we refer to an intensional model.

The starting point for an intensional model is the same as that of an idealized model of gradual typing: namely, a double CBPV model (see Definition 1.2). In particular, unlike the extensional model where we could only compose *relations* horizontally, here we also have a horizontal composition operation on squares.

As in the definition of extensional model, we also require the existence of a natural transformation $\mathcal{U} : 1 \Rightarrow U$ such that $\mathcal{U} \circ ! \sqsubseteq_{r(UB)}^{r(A)} f$ for any $f : A \rightarrow UB$, and a distinguished value type Nat with morphisms $z : \mathcal{V}(1, \text{Nat})$ and $s : \mathcal{V}(\text{Nat}, \text{Nat})$.

We next discuss the aspects that allow us to model intensional behavior. To begin, we need guarded primitives.

1. There is an applicative endofunctor $\triangleright : \mathcal{C} \rightarrow \mathcal{C}$.
2. There is a natural transformation $\text{next} : \text{id} \rightarrow \triangleright$.
3. For any morphism f of the form $\triangleright X \rightarrow X$, there is a unique morphism $\text{fix}f : 1 \rightarrow X$ such that $\text{fix}f = f \circ \text{next} \circ \text{fix}f$.

Notice that by naturality, we have that $(\triangleright g) \circ \text{next} = \text{next} \circ g$ for any $g : X \rightarrow Y$.

We further require there to be a distinguished “think” map $\theta_B : \triangleright(UB) \rightarrow UB$ for all computation types B . From this, we can define the “delay” map $\delta_B := \theta_B \circ \text{next} : UB \rightarrow UB$. Moreover, we require that the think maps are related in that for any $d : B \multimap B'$, we have a square $\theta_B \sqsubseteq_{Ud}^{\triangleright(Ud)} \theta_{B'}$. We also require that these maps commute with computation morphisms, in the sense that for any $\phi \in \mathcal{E}_f(B_1, B_2)$ we have $U\phi \circ \theta_{B_1} = \theta_{B_2} \circ \triangleright(U\phi)$.

We call a model satisfying these requirements an *intensional pre-model*. For the sake of ease of reference, we recap the definition below:

Definition 2.1. An **intensional pre-model** consists of:

- A double CBPV model, i.e., a category internal to the category of CBPV models and lax morphisms, where we require that the morphisms r , s , and t are strict.
- A natural transformation $\mathcal{U} : 1 \Rightarrow U$ such that $\mathcal{U} \circ ! \sqsubseteq_{r(UB)}^{r(A)} f$ for any $f : A \rightarrow UB$.
- A value type Nat with morphisms $z : \mathcal{V}(1, \text{Nat})$ and $s : \mathcal{V}(\text{Nat}, \text{Nat})$.
- A distinguished “think” map $\theta_B \in \mathcal{V}_f(\triangleright(UB), UB)$ for all computation types B . We define $\delta_B := \theta_B \circ \text{next} : \mathcal{V}_f(UB, UB)$.
- For any $d : B \multimap B'$, there is a square $\theta_B \sqsubseteq_{Ud}^{\triangleright(Ud)} \theta_{B'}$.
- For any $\phi \in \mathcal{E}_f(B, B')$ we have $U\phi \circ \theta_{B_1} = \theta_{B_2} \circ \triangleright(U\phi)$.

Lemma 2.1. We have the following properties involving the delay maps in any intensional pre-model:

1. For any $d : B \multimap B'$, we have a square $\delta_B \sqsubseteq_{Ud}^{Ud} \delta_{B'}$.
2. We have $U\phi \circ \delta_{B_1} = \delta_{B_2} \circ U\phi$ for any $\phi \in \mathcal{E}_f(B, B')$.
3. We can define a computation morphism $\delta_{FA}^* \in \mathcal{E}_f(FA, FA)$ for all A . To do so, first compose the unit $\eta_A \in \mathcal{V}_f(A, UFA)$ with the morphism $\delta_{FA} \in \mathcal{V}_f(UFA, UFA)$, and then by the adjunction we get a computation morphism $\delta_{FA}^* \in \mathcal{E}_f(FA, FA)$.
4. There is a square $\delta_{FA}^* \sqsubseteq_{Fc}^{Fc} \delta_{FA'}^*$ for all $c : A \multimap A'$.

Proof. See appendix. □

2.2.1 Weak Bisimilarity

The next ingredient in the definition of intensional model is *weak bisimilarity*, which formalizes the notion that two morphisms are the same “up to steps”, i.e., they differ only in that one may wait more than the other. In particular, for any pair of objects A and A' , in $\mathcal{V}_f$, we require that there is a reflexive, symmetric relation $\approx_{A, A'}$ on the hom-set $\mathcal{V}_f(A, A')$, called the *weak bisimilarity* relation. Similarly for the computation category: there is a reflexive, symmetric relation $\approx_{B, B'}$ defined on each hom-set $\mathcal{E}_f(B, B')$. Additionally, the weak bisimilarity relation should be a congruence with respect to composition: if $f \approx_{A, A'} f'$ and $g \approx_{A', A''} g'$, then $g \circ f \approx_{A, A''} g' \circ f'$, and likewise for computations.

The bisimilarity relation should also be preserved by the functors $U, F, \rightarrow$, and $\times$ e.g., if $f \approx_{A', A} g$ and $\phi \approx_{B, B'} \phi'$, then $f \rightarrow \phi \approx_{A \rightarrow B, A' \rightarrow B'} g \rightarrow \phi'$. Similarly, the two directions of the isomorphism of hom-sets $\mathcal{V}_f(A, UB) \cong \mathcal{E}_f(FA, B)$ should preserve bisimilarity.

We can state all of this compactly by requiring that the categories $\mathcal{V}_f$ and $\mathcal{E}_f$ be *enriched* over the symmetric monoidal category $\mathbf{Set}_\approx$, whose objects are sets equipped with a reflexive, symmetric relation and whose arrows are functions between the sets that preserve the relation. Then we require that the functors $U, F, \rightarrow$, and $\times$ are $\mathbf{Set}_\approx$ -enriched, and likewise $F \dashv U$ as enriched functors. Note that $\mathcal{V}_{sq}$ and $\mathcal{E}_{sq}$ are not $\mathbf{Set}_\approx$ -enriched, so in the functors r, s, t between $\mathcal{V}_f$ and $\mathcal{V}_{sq}$ and between $\mathcal{E}_f$ and $\mathcal{E}_{sq}$, we view $\mathcal{V}_f$ and $\mathcal{E}_f$ as ordinary categories, where we simply forget the bisimilarity relation on hom-sets.

Lastly, we require that for any computation object B , the delay map $\delta_B : UB \rightarrow UB$ (which is defined in terms of θ_B) is bisimilar to the identity: $\delta_B \approx_{UB, UB} \text{id}_{UB}$.

Definition 2.2. An *intensional pre-model with bisimilarity* is an intensional pre-model $\mathcal{M}$ such that:

- The categories $\mathcal{V}_f$ and $\mathcal{E}_f$ are enriched over $\mathbf{Set}_\approx$.
- The functors $U, F, \rightarrow$, and $\times$ are $\mathbf{Set}_\approx$ -enriched, and the adjunction $F \dashv U$ is an enriched adjunction.
- For each B , the delay map δ_B is bisimilar to the identity $\delta_B \approx_{UB, UB} \text{id}_{UB}$.

If $\mathcal{M}$ is an intensional pre-model with bisimilarity, it will sometimes be useful to have a concise definition describing the value objects, morphisms, relations, and squares as a coherent unit, and likewise for the computations. To do this, we introduce some terminology:

Definition 2.3. A *bisimilarity reflexive-graph* $\mathcal{C}$ consists of:

- A category $\mathcal{C}_f$ of objects and vertical morphisms, enriched in $\mathbf{Set}_\approx$.
- A thin category $\mathcal{C}_{sq}$ of horizontal morphisms and squares.
- A functor $r : |\mathcal{C}_f| \rightarrow \mathcal{C}_{sq}$ where $|\mathcal{C}_f|$ is the underlying ordinary category of $\mathcal{C}_f$.
- Functors $s, t : \mathcal{C}_{sq} \rightarrow |\mathcal{C}_f|$ such that $sr = tr = \text{id}$.

A *functor* $G : \mathcal{C} \rightarrow \mathcal{C}'$ of bisimilarity reflexive-graphs consists of:

- A $\mathbf{Set}_\approx$ -enriched functor $G_f : \mathcal{C}_f \rightarrow \mathcal{C}'_f$.
- A functor $G_{sq} : \mathcal{C}_{sq} \rightarrow \mathcal{C}'_{sq}$.
- Source and target are preserved: $s' \circ G_{sq} = G_f \circ s$, and $t' \circ G_{sq} = G_f \circ t$
- Reflexivity is preserved: $r' \circ G_f = G_{sq} \circ r$.

Let $G, H : \mathcal{C} \rightarrow \mathcal{C}'$ be functors of bisimilarity reflexive-graphs. A *natural transformation* $\alpha : G \rightarrow H$ consists of:

- A $\mathbf{Set}_\approx$ -enriched natural transformation $\alpha^f : G_f \rightarrow H_f$.
- A natural transformation $\alpha^{sq} : G_{sq} \rightarrow H_{sq}$.

A *bisimilarity double-category* is a bisimilarity reflexive graph that additionally has a functor $\circ : \mathcal{C}_{sq} \times_{s=t} \mathcal{C}_{sq} \rightarrow \mathcal{C}_{sq}$ where $\mathcal{C}_{sq} \times_{s=t} \mathcal{C}_{sq}$ is the following pullback:

$$\begin{array}{ccc} \mathcal{C}_{sq} \times_{s=t} \mathcal{C}_{sq} & \longrightarrow & \mathcal{C}_{sq} \\ \downarrow & \lrcorner & \downarrow \\ \mathcal{C}_{sq} & \longrightarrow & \mathcal{C}_f \end{array}$$

We additionally require that composition behaves with respect to the source and target functors, and satisfies associativity and left and right unit laws, as in an ordinary double category.

With these definitions in hand, we can give an alternative definition of an intensional pre-model with bisimilarity:

Definition 2.4.

2.2.2 Perturbations

We next focus on the specification of perturbations in a pre-model with bisimilarity. To begin, let $\mathcal{C}$ be a bisimilarity double-category. For an object X of $\mathcal{C}$, a *semantic perturbation* of X is an endomorphism $f \in \mathcal{C}_f(X, X)$ such that $f \approx_{X, X} \text{id}_X$. The collection of semantic perturbations on X forms a monoid under composition, with the identity element being the identity id_X . We write $\text{Endo}_{\approx}(X)$ to refer to this monoid.

For every object $X \in \mathcal{C}_f$, we require that there is a commutative monoid² M_X of *syntactic perturbations* of X and a monoid homomorphism $i_X : M_X \rightarrow \text{Endo}_{\approx}(X)$ that interprets these as semantic perturbations.

The prototypical semantic perturbation for a computation object B is the delay map $\delta_B : UB \rightarrow UB$. We require that for any computation object B , the monoid M_{UB} contains an element m such that $i_{UB}(m) = \delta_B$.

We require that the functors $\times, \overset{k}{\times}, \rightarrow, \overset{k}{\rightarrow}, U, \overset{k}{U}, F, \overset{k}{F}$ all preserve perturbations. As in the prior work of Giovannini et al., we also require that the perturbations interact nicely with the relation morphisms. Namely, if $R : X \multimap X'$ is a relation morphism, we require there to be two homomorphisms of monoids $\text{push}_R : M_X \rightarrow M_{X'}$ and $\text{pull}_R : M_{X'} \rightarrow M_X$. The authors call this the *push-pull property*.

If $\mathcal{M}$ is a pre-model with bisimilarity that additionally satisfies the above properties, we say that $\mathcal{M}$ is a *pre-model with perturbations*. To summarize:

Definition 2.5. A *pre-model with perturbations* $\mathcal{M}$ is a pre-model with bisimilarity, such that the following additional requirements are met:

1. For each object (value or computation) X , there is a commutative monoid M_X and a homomorphism of monoids $i_X : M_X \rightarrow \text{Endo}_{\approx}(X)$.
2. For all B , there is an element m such that $i_{UB}(m) = \delta_B$.
3. The functors $\times, \overset{k}{\times}, \rightarrow, \overset{k}{\rightarrow}, U, \overset{k}{U}, F, \overset{k}{F}$ preserve perturbations.
4. The push-pull property holds for all $c : A \multimap A'$ and all $d : B \multimap B'$.

2.3 Representable Relations

Now we turn to the notion of representability of relations. Here we make a departure from prior work: rather than working with quasi-representability, we instead work with *actual* representability in a certain double category of *well-behaved morphisms*. The idea is as follows. Recall that in the model of Giovannini et al., the vertical morphisms had no interaction with the monoid of perturbations; using their terminology, a morphism of value objects (predomains equipped with syntactic perturbations) was simply a morphism of the underlying predomains. We observe, however, that many morphisms actually *do* interact nicely with perturbations. In particular, the cast morphisms enjoy this property. (The precise sense in which we mean “interact nicely” will be explained below.) This observation leads to the following idea. If we define a notion of square where the sides build in a perturbation, we can look at representable relations in this double category rather than defining a separate, ad-hoc notion of quasi-representable relations. The interaction between morphisms and perturbations turns out to be precisely what we need for these squares to compose, i.e., for these squares to form a double category.

More concretely, given a pre-model with perturbations $\mathcal{M}$, we will define a double category $\mathcal{V}^*$ whose vertical morphisms consist of value morphisms that behave well with respect to casts, and whose squares are squares in which both sides are composed with perturbations. Likewise, we can define analogously a double category $\mathcal{E}^*$. In particular, let $\mathcal{C}$ stand for the double category of values of $\mathcal{M}$, or the double category of computations of $\mathcal{M}$. We first define what we mean for a vertical morphism to have an action on perturbations:

Definition 2.6. Let $f \in \mathcal{C}(X_i, X_o)$. We say that f *commutes with perturbations* if:

- There are monoid homomorphisms $f^{\rightarrow} : M_{X_i} \rightarrow M_{X_o}$ and $f^{\leftarrow} : M_{X_o} \rightarrow M_{X_i}$.

²The model of Giovannini et al. used (not necessarily commutative) monoids, whereas we demand that the monoids be commutative. The reason for this will become clear in the next section.

- For all $m_i \in M_{X_i}$ and $m_o \in M_{X_o}$ we have

$$(i(m_o) \circ f) \sqsubseteq_{r(X_o)}^{r(X_i)} (f \circ i(f^{\leftarrow}(m_o)))$$

and

$$(i(f^{\rightarrow}(m_i)) \circ f) \sqsubseteq_{r(X_o)}^{r(X_i)} (f \circ i(m_i)).$$

Remark 1. Not all vertical morphisms in $\mathcal{M}$ commute with perturbations.

Remark 2. The reason for ordering rather than equality is that semantic perturbations for computation types are computation morphisms, and hence need to commute with the algebraic structure. Therefore, performing the perturbation after performing a computation morphism may not result in any delay being inserted, while performing the perturbation first will always insert a delay. At first glance this would seem to be problematic, since one side has a delay where the other does not. However, the key idea is that the vertical morphisms are meant to be casts, which have no effects other than to delay or to error. In either case, we will be able to show that the side that performs the perturbation after the morphism is *below* the side that performs the perturbation first.

A square in this double category will be defined as a square in the usual sense, but where we are allowed to synchronize the vertical morphisms on either side with a perturbation. More concretely:

Definition 2.7. Let $f : X_i \rightarrow X_o$ and $g : X'_i \rightarrow X'_o$ be perturbation-preserving morphisms and let $R_i : X_i \multimap X'_i$ and $R_o : X_o \multimap X'_o$ be horizontal morphisms. We say that there is a **perturbed square** between f and g , written $f \sqsubseteq_{R_o}^{\text{ptb}} g$, if there *merely exist* perturbations m_f, m'_f, m_g, m'_g such that $i(m'_f) \circ f \circ i(m_f) \sqsubseteq_{R_o}^{R_i} i(m'_g) \circ g \circ i(m_g)$.

$$\begin{array}{ccc}
 X_i & \xrightarrow{R_i} & X'_i \\
 f \downarrow & & \downarrow g \\
 X_o & \xrightarrow{R_o} & X'_o
 \end{array}
 \rightsquigarrow
 \begin{array}{ccc}
 X_i & \xrightarrow{R_i} & X'_i \\
 i(m_f) \downarrow & & \downarrow i(m_g) \\
 X_i & & X'_i \\
 f \downarrow & & \downarrow g \\
 X_o & & X'_o \\
 i(m'_f) \downarrow & & \downarrow i(m'_g) \\
 X_o & \xrightarrow{R_o} & X'_o
 \end{array}$$

Perturbed squares can be composed vertically and horizontally, as shown in the following two lemmas:

Lemma 2.2. Suppose $f_1 \sqsubseteq_{R_2}^{\text{ptb}} g_1$ and $f_2 \sqsubseteq_{R_3}^{\text{ptb}} g_2$. Then $(f_2 \circ f_1) \sqsubseteq_{R_3}^{\text{ptb}} (g_2 \circ g_1)$.

Lemma 2.3. If $f \sqsubseteq_{R_o}^{\text{ptb}} g$ and $g \sqsubseteq_{R'_o}^{\text{ptb}} h$, then $f \sqsubseteq_{R_o \odot R'_o}^{\text{ptb}} h$.

Vertical composition uses the fact that the vertical morphisms preserve perturbations, while the horizontal composition uses the fact that the monoids are commutative, as well as the push-pull property of the relations. For details, see the proofs in the appendix.

With these definitions, we define a locally-thin double category $\mathcal{C}^*$ of morphisms commuting with perturbations:

Definition 2.8. Let $\mathcal{M}$ be an intensional model with perturbations, and let $\mathcal{C}$ be a bisimilarity double-category. We define a new double category $\mathcal{C}^*$ as follows:

- The objects of $\mathcal{C}^*$ are the same as those of $\mathcal{C}$.
- The vertical morphisms of $\mathcal{C}^*$ are those morphisms of $\mathcal{C}$ that commute with perturbations.

- The horizontal morphisms of $\mathcal{C}^*$ are those of $\mathcal{C}$.
- The squares of $\mathcal{C}^*$ are perturbed squares.

Note that this is locally-thin because the perturbations in the definition of perturbed square are existentially quantified. We can now define a notion of an intensional model where the horizontal morphisms are representable:

Definition 2.9. Let $\mathcal{M}$ be an intensional pre-model with perturbations. We say that $\mathcal{M}$ is a *pre-model with representability* if:

- Every value horizontal morphism $c : A \multimap A'$ is left-representable in the double category $\mathcal{V}^*$. That is, there is a perturbation-preserving morphism $\text{up } c \in \mathcal{V}^*(A, A')$ that left-represents c .
- Every computation horizontal morphism $d : B \multimap B'$ is right-representable in $\mathcal{E}^*$. That is, there is a perturbation-preserving morphism $\text{dn } d \in \mathcal{E}^*(B', B)$ that right-represents d .

2.4 The Dynamic Type

We now discuss how to specify the denotation of the dynamic type in an intensional model. It is important to note that it makes sense to axiomatize the dynamic type already in a base intensional pre-model. That is, we do not require the notions of weak bisimilarity, perturbations, or representability of horizontal morphisms to define the denotation of the dynamic type.

Let $\mathcal{M}$ be an intensional pre-model, a pre-model with bisimilarity, a pre-model with perturbations, or a pre-model with representability. To interpret the dynamic type, we require that there is a distinguished value object $D \in \mathcal{V}$, such that for every value object A , there is a distinguished horizontal morphism $\text{inj}_A : A \multimap D$.

Now we can state the definition of an intensional model:

Definition 2.10. An *intensional model of gradual typing* is an intensional pre-model with representability that furthermore has an interpretation of the dynamic type.

3 Constructing a Guarded Model of Gradual Typing

In this section, we discuss how to *construct* an abstract model of gradual typing via a step-by-step process. Recall that the ultimate goal is to build an extensional model of gradual typing (Definition ??), as such a model contains the structure necessary to interpret the syntax and precision rules of a gradual cast calculus.

Although the definition of an extensional model involves a significant amount of data, it is not necessary to create an instance of an extensional model from scratch. Much of the difficult work can be carried out abstractly. In particular, it is possible to construct an extensional model from an intensional model using the abstract analogue of the bisimilarity-closure construction in the work of Giovannini et al. Moreover, much of the construction of an intensional model can also be carried out abstractly.

3.1 Starting Point: Intensional Pre-Model with Bisimilarity

We take as a starting point an intensional pre-model with bisimilarity (Definition 2.2).

3.2 Adding Perturbations

Suppose we are given an intensional pre-model with bisimilarity $\mathcal{M}$. Our goal is to construct a model $\mathcal{M}'$ with perturbations, as in Definition 2.5. Furthermore, the vertical morphisms in $\mathcal{M}'$ will commute with perturbations, as defined in Definition 2.6.

The summary of the construction is as follows: a value object in $\mathcal{M}'$ consists of a value object A of $\mathcal{M}$ paired with a commutative monoid M_A of syntactic perturbations, and a homomorphism of monoids $i_A : M_A \rightarrow \text{Endo}_{\approx}(A)$ interpreting the perturbations as semantic perturbations. For reasons that will become clear

when we define the functor F , we also add a second commutative monoid of *impure* or *Kleisli* perturbations M_A^k along with a monoid homomorphism $i_A^k : M_A^k \rightarrow \text{Endo}_{\approx}(FA)$. Dually, a computation object of $\mathcal{M}'$ consists of a computation object B of $\mathcal{M}$ paired with commutative monoids M_B and M_B^k , along with monoid homomorphisms $i_B : M_B \rightarrow \text{Endo}_{\approx}(B)$ and $i_B^k : M_B^k \rightarrow \text{Endo}_{\approx}(UB)$.

A value morphism $f \in \mathcal{V}'_f(A_i \rightarrow A_o)$ in $\mathcal{M}'$ is defined to be a value morphism in $\mathcal{M}$ such that f commutes with the perturbations in M_{A_i} and M_{A_o} , and Ff commutes with the perturbations in $M_{A_i}^k$ and $M_{A_o}^k$. Dually, a computation morphism $\phi \in \mathcal{E}'_f(B_i, B_o)$ is defined to be a computation morphism in $\mathcal{M}$ that commutes with the perturbations in M_{B_i} and M_{B_o} , and such that $U\phi$ commutes with the perturbations in $M_{B_i}^k$ and $M_{B_o}^k$.

A value relation $c : A \multimap A'$ in $\mathcal{M}'$ is defined to be a value relation in $\mathcal{M}$ that satisfies the push-pull property with respect to the monoids M_A and $M_{A'}$, and such that Fc satisfies push-pull with respect to M_A^k and $M_{A'}^k$. Dually, a computation relation $d : B \multimap B'$ in $\mathcal{M}'$ is defined to be a computation relation in $\mathcal{M}$ that satisfies the push-pull property with respect to M_B and $M_{B'}$, and such that Ud satisfies push-pull with respect to M_B^k and $M_{B'}^k$.

Value and computation squares in $\mathcal{M}'$ are defined to be the same as those in $\mathcal{M}$.

Next, recall from Section ?? that we can construct a bisimilarity double-category whose vertical morphisms commute with perturbations and whose squares involve pre- and post-composing each vertical morphism with a perturbation.

3.3 Adding Representability

3.4 The Dynamic Type

3.5 Extracting an Extensional Model: The Bisimilarity-Closure Construction

Lastly, we can obtain an extensional model from an intensional model using the abstract analogue of the bisimilarity-closure definition in the work of Giovannini et al. In particular, let $\mathcal{M}$ be an intensional model. We define an extensional model $\mathcal{M}'$ as follows. The objects and vertical/horizontal morphisms are the same as those in $\mathcal{M}$; it is only the definition of squares that differs.

Specifically, let $f : X_i \rightarrow X_o$ and $g : X'_i \rightarrow X'_o$ be vertical morphisms (of either value or computation types). Let $R_i : X_i \multimap X'_i$ and $R_o : X_o \multimap X'_o$ be horizontal morphisms. We define

$$f \overset{R_i}{\sqsubseteq}_{R_o} g := \exists f', g'. f \approx_{X_i, X_o} f' \sqsubseteq_{R_o}^{R_i} g' \approx_{X'_i, X'_o} g.$$

It remains to verify that the model satisfies the requirements of an extensional model. The basic idea is that the perturbations in the intensional squares are all bisimilar to the identity function, and hence they disappear when we perform the above construction.